

MAT11 - 1

Foundations

[Module page](#)

Version of 13 September 2026

About this textbook

This year, for the first time, every Bachelor course at Albert School has its own textbook, thanks to your teachers and AI.

- The textbook is a help for working on the course at home, after the class.
- The course as taught in class remains the reference, and prevails over the textbook wherever the two differ.
- This is a first version. Help us improve it by sharing your feedback with your teacher.

Contents

Preface	5
1. The Language of Logic	6
1.1. Reading a statement and its converse	6
1.2. Propositions and connectives	6
1.3. Contrapositive, converse, and the mistakes they cause	8
1.4. De Morgan's laws and distributivity	8
1.5. Quantifiers	9
2. Sets	10
2.1. Collecting objects precisely	10
2.2. Membership, inclusion, and description	10
2.3. Operations on sets	10
2.4. Power set	11
2.5. Cartesian product	11
3. Counting: Combinatorics	13
3.1. Why counting needs a method	13
3.2. Factorial and permutations	13
3.3. Arrangements and combinations	13
3.4. Pascal's rule and Pascal's triangle	14
4. Methods of Proof	16
4.1. Templates worth memorising	16
4.2. Direct proof and proof by contrapositive	16
4.3. Proof by exhaustion	16
4.4. Proof by contradiction	17
4.5. Proof by induction	17
4.6. Analysis–synthesis	18
5. Sums and Products	20
5.1. A compact notation for long sums	20
5.2. Three rules for rearranging a sum	20
5.3. The same three moves for products	21
5.4. The standard sums	22
5.5. The geometric sum	22
5.6. The binomial theorem	23
6. Functions	25
6.1. A machine, then a definition	25
6.2. Composition	25
6.3. Image and preimage of a set	26
6.4. Injective, surjective, bijective	26
6.5. Inverse functions	27
6.6. The bijection theorem	28
7. Comparing Infinities	29
7.1. Comparing sizes by pairing	29
7.2. Equipotence and countable sets	29
7.3. Cantor's theorem	29
7.4. The Cantor–Bernstein theorem	30
8. Sequences	31

8.1.	Real sequences	31
8.2.	Convergence	31
8.3.	Operations on limits	32
8.4.	The squeeze theorem and monotone convergence	33
8.5.	Arithmetico-geometric sequences	33
9.	Differentiation	35
9.1.	Working near a point	35
9.2.	Limits and continuity	35
9.3.	The derivative	36
9.4.	Differentiability and continuity	37
9.5.	The differentiation rules	38
9.6.	Monotonicity from the sign of the derivative	38
9.7.	Local extrema and Fermat's condition	39
9.8.	The second derivative: convexity and concavity	40
10.	Integration	42
10.1.	Signed area and antiderivatives	42
10.2.	Recognising usual forms	42
10.3.	The Fundamental Theorem of Calculus	43
10.4.	Linearity, Chasles, and the bounds	43
10.5.	Integration by parts	44
10.6.	Substitution	45
	Notation reference	46
11.	Appendix: derivatives and antiderivatives	47
11.1.	Standard derivatives	47
11.2.	Composite derivatives	47
11.3.	Standard antiderivatives	48
11.4.	Composite antiderivatives	48

Preface

By the end of this course you will be able to do a set of things you cannot do yet. You will **read** a mathematical statement — decide exactly what it claims, form its negation, and tell it apart from its converse. You will **prove** a statement by each of the standard templates: directly, by contrapositive, by contradiction, by exhaustion of cases, by induction, and by analysis–synthesis. You will speak the language of **sets** and **logic**, **count** finite collections with permutations and combinations, and **manipulate sums and products** in index notation, including the binomial theorem and the standard closed forms. You will classify a **function** as injective, surjective, or bijective, invert it, and work with images and preimages; and you will **compare infinite sets**, telling a countable one from an uncountable one. You will decide whether a **sequence** converges and compute its limit. Finally, you will **differentiate** a function, read its variations and extrema, study its convexity, and **integrate** it — using antiderivatives, the Fundamental Theorem of Calculus, integration by parts, and substitution.

1. The Language of Logic

1.1. Reading a statement and its converse

Consider the claim: “**if a whole number n is divisible by 6, then it is divisible by 3.**” You believe it. Now consider its **converse**: “**if n is divisible by 3, then it is divisible by 6.**” That is false ($n = 9$). Same two properties, the arrow reversed, and truth flips to falsehood. Many reasoning errors in mathematics are exactly this: confusing a statement with one of its rearrangements. This chapter builds the machinery that makes such confusions impossible.

1.2. Propositions and connectives

Definition 1 (Proposition). A **proposition** is a statement that is either **true** or **false**, but not both. “ $5 > 3$ ” is a proposition (true). “ $x > 3$ ” is **not** a proposition until we know x — it is a **predicate**, a statement with a free variable, which becomes a proposition once the variable is fixed or quantified.

From propositions P and Q we build compound propositions using **connectives**. Each is defined by a **truth table**, which lists the truth value of the compound for every combination of inputs.

Definition 2 (The basic connectives).

- **Negation** $\neg P$ (“not P ”) is true exactly when P is false.
- **Conjunction** $P \wedge Q$ (“ P and Q ”) is true exactly when both are true.
- **Disjunction** $P \vee Q$ (“ P or Q ”) is true when at least one is true. This is the **inclusive** or: it is **true** when both hold.
- **Implication** $P \implies Q$ (“ P implies Q ”, “if P then Q ”) is false **only** in the single case P true and Q false.
- **Equivalence** $P \iff Q$ (“ P if and only if Q ”) is true exactly when P and Q have the same truth value.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \implies Q$	$P \iff Q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Table 1: The truth tables of the five connectives on one grid. Read each row as one possible situation; each column tells you whether the compound is true in that situation.

Definition 3 (Truth table, logical equivalence). The **truth table** of a compound proposition lists its truth value for every combination of truth values of the atomic propositions it is built from — 2^k rows for k atoms. Two propositions are **logically equivalent**, written $P \equiv Q$, when they have the same truth table: the same truth value in every row.

The implication row is worth examining closely. Why is $P \implies Q$ **true** whenever P is false? Think of a promise: “**if it rains, I will bring an umbrella.**” You have broken that promise only on a day when it rains and you have no umbrella. On a dry day you have broken nothing — whatever you carry. An implication with a false hypothesis is **vacuously true**; it makes no claim. And because a false hypothesis makes **every** implication true, from a contradiction $P \wedge \neg P$ one could infer any proposition at all — the **principle of explosion**, which is exactly what makes reaching a contradiction fatal to an assumption (Chapter 4).

Pitfall. $P \implies Q$ does **not** mean that P causes Q , nor that P and Q are both true. “If $2 + 2 = 5$, then 7 is even” is a **true** implication, because its hypothesis is false — even though its conclusion is false too. Implication is about the **pattern of truth values**, not about meaning or cause.

An implication is also the standard way to phrase a **condition**.

Definition 4 (Necessary and sufficient conditions). For the implication $P \implies Q$:

- P is a **sufficient condition** for Q — the truth of P suffices to guarantee Q ;
- Q is a **necessary condition** for P — Q must hold whenever P does.

An implication thus delivers a **sufficient** condition for its conclusion, and equivalently a **necessary** condition for its hypothesis. An equivalence $P \iff Q$ makes each of P and Q both necessary and sufficient for the other.

To analyse a **compound** proposition built from several connectives, extend the table one column at a time, working from the innermost pieces outward. Each new column is computed from columns already filled, using the row-by-row rules above.

Example — An implication rewritten without the arrow. Is $P \implies Q$ the same as $(\neg P) \vee Q$? Build both, one column at a time. There are two variables, so $2^2 = 4$ rows. Add a column for $\neg P$, combine it with Q using the “or” rule, then compare with the implication column.

P	Q	$\neg P$	$(\neg P) \vee Q$	$P \implies Q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

The last two columns are identical, so $P \implies Q \equiv (\neg P) \vee Q$: an implication **is** “not the hypothesis, or the conclusion”. In plain words, the promise “if it rains, I bring an umbrella” says exactly “it is not raining, or I bring an umbrella” — the two describe the very same set of days.

Three further laws govern negation itself.

Proposition 1 (Excluded middle, non-contradiction, involution). For every proposition P , the proposition $P \vee \neg P$ is always true (the **law of the excluded middle**), and $P \wedge \neg P$ is always false (**non-contradiction**); and negating twice changes nothing (**involution of negation**):

$$\neg(\neg P) \equiv P.$$

1.3. Contrapositive, converse, and the mistakes they cause

Given $P \implies Q$, three related implications can be formed.

Definition 5 (Converse, inverse, contrapositive). For the implication $P \implies Q$:

- its **converse** is $Q \implies P$;
- its **inverse** is $\neg P \implies \neg Q$;
- its **contrapositive** is $\neg Q \implies \neg P$.

The single most useful fact in this chapter:

Theorem 1 (An implication equals its contrapositive). $P \implies Q$ and $\neg Q \implies \neg P$ are **logically equivalent** — they have identical truth tables. By contrast, the converse and inverse are **not** equivalent to the original, though they are equivalent to each other.

Proof. Compare columns. $P \implies Q$ is false only when P is T and Q is F. Now $\neg Q \implies \neg P$ is false only when $\neg Q$ is T and $\neg P$ is F, that is, Q is F and P is T — the very same situation. Two propositions false in exactly the same rows are true in exactly the same rows, hence equivalent. $\square$

Pitfall. Affirming the converse is a classic error. From “if it rained, the ground is wet” and “the ground is wet”, you may **not** conclude “it rained” — someone may have used a hose. The truth of $P \implies Q$ tells you nothing about $Q \implies P$.

1.4. De Morgan’s laws and distributivity

To disprove a claim, or to run a proof by contradiction, you must negate it correctly. The rules are mechanical once learned.

Proposition 2 (De Morgan’s laws and the negation of an implication).

$$\begin{aligned}\neg(P \wedge Q) &\equiv (\neg P) \vee (\neg Q), \\ \neg(P \vee Q) &\equiv (\neg P) \wedge (\neg Q), \\ \neg(P \implies Q) &\equiv P \wedge (\neg Q).\end{aligned}$$

The last one is the most often mishandled. The negation of “if P then Q ” is **not** another implication — it is the assertion that P holds **and yet** Q fails. To deny the promise “if it rains I bring an umbrella” is to describe one rainy, umbrella-less day.

The connectives $\wedge$ and $\vee$ also **distribute** over one another, exactly as $\times$ distributes over $+$ in arithmetic.

Proposition 3 (Distributivity of conjunction and disjunction).

$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R),$$

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R).$$

Both are verified by an eight-row truth table, one row per truth assignment to P, Q, R . Distributivity is the logical backbone of the set identities in Chapter 2: union distributes over intersection precisely because “or” distributes over “and”.

1.5. Quantifiers

So far a predicate such as “ $x > 3$ ” became a proposition only once x was fixed. **Quantifiers** turn a predicate into a proposition in a second way: by stating for how many values of the variable it holds.

Definition 6 (Universal and existential quantifiers). For a predicate $P(x)$ ranging over a set of values:

- $\forall x, P(x)$: “**for all** x , $P(x)$ holds”;
- $\exists x, P(x)$: “**there exists** an x with $P(x)$ ”;
- $\exists! x, P(x)$: “there exists a **unique** such x ”.

Pitfall. The order of quantifiers changes the meaning. Over the whole numbers, “ $\forall x, \exists y : y > x$ ” (every number has a larger one — true) is utterly different from “ $\exists y, \forall x : y > x$ ” (one number is larger than all — false). When you swap $\forall$ and $\exists$ you have written a new statement, usually with a new truth value.

Of the two orders, the one with $\exists$ **before** $\forall$ is the **stronger**: a single y that works for every x is a more demanding requirement. Indeed “ $\exists y, \forall x : P(x, y)$ ” implies “ $\forall x, \exists y : P(x, y)$ ” — the same y serves each x — while the converse fails, as the example above shows.

To negate a quantified statement, push the negation inward, flipping each quantifier as it passes:

$$\neg(\forall x, P(x)) \equiv \exists x, \neg P(x),$$

$$\neg(\exists x, P(x)) \equiv \forall x, \neg P(x).$$

“Not everyone passed” means “someone did not pass” — not “everyone failed”. A chain of quantifiers is negated left to right: every $\forall$ becomes $\exists$, every $\exists$ becomes $\forall$, and the predicate at the end is negated.

Example — Negating a two-quantifier statement. Negate “ $\forall \varepsilon > 0, \exists n : 1/n < \varepsilon$ ”. Flip each quantifier and negate the inside:

$$\exists \varepsilon > 0, \forall n : 1/n \geq \varepsilon.$$

The original says “ $1/n$ can be made as small as we like”; its negation claims a positive ε that $1/n$ never drops below. The original is true, so its negation is false — but negating correctly is a separate skill from deciding truth.

2. Sets

2.1. Collecting objects precisely

A theorem is only as clear as the objects it speaks of. “The whole numbers between 1 and 10 with no odd divisor above 1” is a sentence; $\{2, 4, 8\}$ is a set you can compute with. A **set** is the basic container of mathematics, and every operation on sets, as we will see, is a logical connective from Chapter 1 applied to the statement “ x belongs”.

2.2. Membership, inclusion, and description

Definition 7 (Set, membership, inclusion). A **set** is a collection of distinct objects, its **elements**. We write:

- $x \in A$ for “ x is an element of A ”, and $x \notin A$ to deny it;
- $A \subset B$ (“ A is a subset of B ”) when every element of A is an element of B , that is, $x \in A \implies x \in B$;
- $A = B$ when $A \subset B$ and $B \subset A$ — the two sets have exactly the same elements.

The **empty set** $\emptyset$ has no elements; it is a subset of every set.

A set may be given by **extension**, listing its elements, $\{2, 4, 8\}$, or by **comprehension** $\{x \in A : P(x)\}$, read “the elements x of A such that $P(x)$ holds”. Two facts about listing: order does not matter and repetition does not count, so $\{1, 2, 2\} = \{2, 1\}$; a one-element set such as $\{2\}$ is a **singleton**. The standard number systems, nested by inclusion, are

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R},$$

the naturals (whole numbers 0, 1, 2, ...), integers, rationals, and reals.

Pitfall. Membership and inclusion are different relations. An element belongs ($2 \in \{1, 2, 3\}$); a set is included ($\{2\} \subset \{1, 2, 3\}$). Writing $2 \subset \{1, 2, 3\}$ or $\{2\} \in \{1, 2, 3\}$ confuses the two. The set $\{2\}$ — a box containing the number 2 — is not the number 2.

Definition 8 (Cardinality). A set is **finite** if it has finitely many elements. Its **cardinality** $\text{Card}(A)$ (also written $|A|$) is that number of elements. For example $\text{Card}(\emptyset) = 0$ and $\text{Card}(\{2, 4, 8\}) = 3$.

2.3. Operations on sets

Definition 9 (Union, intersection, difference, complement). For sets A, B inside a fixed universe E :

- **union** $A \cup B = \{x : x \in A \vee x \in B\}$ — in either;
- **intersection** $A \cap B = \{x : x \in A \wedge x \in B\}$ — in both;

- **difference** $A \setminus B = \{x : x \in A \wedge x \notin B\}$ — in A but not in B ;
- **complement** $A^c = E \setminus A = \{x \in E : x \notin A\}$ — everything in the universe outside A .

Each operation is a connective applied to “ x belongs”: union is “or”, intersection is “and”, complement is “not”. This is why the logic of Chapter 1 transfers wholesale to sets. De Morgan’s laws, read through membership, become the set identities

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c,$$

and distributivity of $\wedge$ over $\vee$ becomes $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. The complement obeys three further laws, direct from its definition:

$$(A^c)^c = A, \quad A \cap A^c = \emptyset, \quad A \cup A^c = E.$$

Example — Computing with a concrete universe. Let $E = \{1, 2, \dots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$. Then $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, $A \cap B = \{4, 5\}$, $A \setminus B = \{1, 2, 3\}$, and $A^c = \{6, 7, 8, 9, 10\}$. Notice $\text{Card}(A \cup B) = 7 = 5 + 4 - 2 = \text{Card}(A) + \text{Card}(B) - \text{Card}(A \cap B)$ — the doubly counted overlap is subtracted once. This is the **inclusion–exclusion** count for two sets.

2.4. Power set

Definition 10 (Power set). The **power set** $\mathcal{P}(A)$ of a set A is the set whose elements are **all** the subsets of A , including $\emptyset$ and A itself.

Example — Listing a power set. For $A = \{1, 2, 3\}$,

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\},$$

which has $8 = 2^3$ elements. That the count is a power of two is no accident; we prove $\text{Card}(\mathcal{P}(A)) = 2^{\text{Card}(A)}$ in Chapter 4, once induction is available.

Pitfall. The elements of $\mathcal{P}(A)$ are **sets**, not the elements of A . So $2 \notin \mathcal{P}(\{1, 2, 3\})$, but $\{2\} \in \mathcal{P}(\{1, 2, 3\})$ and $\{2\} \subset \{1, 2, 3\}$. Belonging to the power set **is** being a subset.

2.5. Cartesian product

Definition 11 (Ordered pair and Cartesian product). An **ordered pair** (a, b) records two objects **in order**: $(a, b) = (c, d)$ exactly when $a = c$ and $b = d$. The **Cartesian product** of A and B is

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\},$$

the set of all such pairs. For finite sets $\text{Card}(A \times B) = \text{Card}(A) \cdot \text{Card}(B)$.

Unlike a set, an ordered pair is sensitive to order: $(1, 2) \neq (2, 1)$, whereas $\{1, 2\} = \{2, 1\}$. The plane $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ is the most familiar Cartesian product — a point is an ordered pair of coordinates.

Example — A small product. With $A = \{1, 2\}$ and $B = \{x, y, z\}$,

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\},$$

which has $2 \times 3 = 6$ elements, matching the cardinality rule.

3. Counting: Combinatorics

3.1. Why counting needs a method

How many ways can eight runners finish a race? How many five-card hands are there? Listing is hopeless once the numbers grow, so we need **rules** that build large counts from small ones. Two rules generate everything in this chapter. The **product rule**: if a first choice can be made in m ways and, for each, a second in n ways, the pair can be made in mn ways. The **sum rule**: if two cases are mutually exclusive, the number of outcomes across both is the sum of the two counts.

3.2. Factorial and permutations

Definition 12 (Factorial). For a whole number n , the **factorial** is

$$n! = n \times (n - 1) \times \cdots \times 2 \times 1,$$

the product of the integers from 1 to n , with the convention $0! = 1$ (the empty product). Thus $1! = 1$, $2! = 2$, $3! = 6$, $4! = 24$, $5! = 120$.

Definition 13 (Permutation). A **permutation** of a finite set is an ordering of all its elements in a row.

Proposition 4 (Number of permutations). A set of n distinct elements has exactly $n!$ permutations.

Proof. Fill n positions left to right. The first position can be any of the n elements; the second, any of the remaining $n - 1$; and so on down to 1 choice for the last. By the product rule the number of orderings is $n \times (n - 1) \times \cdots \times 1 = n!$. $\square$

Example — Ordering runners. Eight runners can finish in $8! = 40320$ different orders. If we only record the first three places, we are counting **arrangements**, the next idea.

3.3. Arrangements and combinations

From eight runners, how many gold–silver–bronze podiums are possible, and how many three-runner committees? Both select three runners from eight, but the podium distinguishes the order of the three while the committee does not. Ordered selections are **arrangements**; unordered ones are **combinations**.

Definition 14 (Arrangement). An **arrangement** of k elements from a set of n (a k -permutation) is an ordered selection of k **distinct** elements. Their number is

$$A_n^k = n \times (n-1) \times \cdots \times (n-k+1) = \frac{n!}{(n-k)!}.$$

The formula is the product rule again: n choices for the first slot, $n-1$ for the second, down to $n-k+1$ for the k -th — a product of k descending factors.

Definition 15 (Combination and binomial coefficient). A **combination** of k elements from a set of n is an **unordered** selection of k distinct elements — that is, a k -element subset. Their number is the **binomial coefficient**

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{A_n^k}{k!}, \quad \text{for } 0 \leq k \leq n,$$

and $\binom{n}{k} = 0$ when $k > n$ or $k < 0$.

The division by $k!$ is the crux: each unordered subset of size k can be ordered in $k!$ ways, so arrangements overcount combinations by exactly the factor $k!$. Dividing removes the order.

Pitfall. Order in, order out. Use A_n^k when the order of the selection matters (a podium: gold, silver, bronze are distinct roles); use $\binom{n}{k}$ when it does not (a committee of three: no roles). Choosing the wrong one is the most common counting error. A quick check: $A_n^k = k! \binom{n}{k}$, so arrangements always exceed combinations by the factor $k!$.

Example — Podium versus committee. From 8 runners, the number of possible gold–silver–bronze podiums is $A_8^3 = 8 \times 7 \times 6 = 336$, whereas the number of three-runner committees is $\binom{8}{3} = 336/3! = 336/6 = 56$. Same eight people, same three chosen — but the podium distinguishes the order and the committee does not.

3.4. Pascal's rule and Pascal's triangle

Binomial coefficients satisfy one recurrence that generates them all without a single factorial.

Theorem 2 (Pascal's rule — milestone proof). For all integers $n \geq 1$ and $k \in \llbracket 1, n \rrbracket$,

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Proof. We count the k -element subsets of $\{1, 2, \dots, n\}$ in two ways. On one hand there are $\binom{n}{k}$ of them, by definition.

On the other hand, split these subsets according to whether they contain the element n — two mutually exclusive, exhaustive cases.

- Subsets **not** containing n are exactly the k -element subsets of $\{1, \dots, n-1\}$: there are $\binom{n-1}{k}$ of them.
- Subsets **containing** n are formed by adjoining n to a $(k-1)$ -element subset of $\{1, \dots, n-1\}$; that smaller subset can be chosen in $\binom{n-1}{k-1}$ ways, and each choice gives a distinct subset containing n .

By the sum rule the total is $\binom{n-1}{k} + \binom{n-1}{k-1}$. Counting the same collection two ways, the two totals are equal. $\square$

This recurrence, together with the border values $\binom{n}{0} = \binom{n}{n} = 1$, lets us build **Pascal's triangle**: each entry is the sum of the two entries above it.

$$\begin{array}{cccccccc} n=0 & 1 & & & & & & \\ n=1 & 1 & 1 & & & & & \\ n=2 & 1 & 2 & 1 & & & & \\ n=3 & 1 & 3 & 3 & 1 & & & \\ n=4 & 1 & 4 & 6 & 4 & 1 & & \\ n=5 & 1 & 5 & 10 & 10 & 5 & 1 & \\ n=6 & 1 & 6 & 15 & 20 & 15 & 6 & 1 \end{array}$$

Table 2: Pascal’s triangle: row n lists $\binom{n}{0}, \dots, \binom{n}{n}$. Each inner entry is the sum of the two just above it (for instance $15 = 5 + 10$), which is exactly Pascal’s rule. Each row is symmetric, since $\binom{n}{k} = \binom{n}{n-k}$.

The symmetry $\binom{n}{k} = \binom{n}{n-k}$ visible in every row has a one-line reason: choosing the k elements to include is the same as choosing the $n - k$ elements to leave out.

4. Methods of Proof

4.1. Templates worth memorising

Up to now we have proved things as the need arose. This chapter isolates the standard **proof templates** the course relies on. Each has a fixed shape you can reproduce: recognise which template a claim invites, and half the work is done. The **direct/contrapositive** pair and **contradiction** dispose of a single implication; **exhaustion** splits a claim into finitely many cases; **induction** proves a statement for every whole number; and **analysis–synthesis** settles existence and uniqueness.

4.2. Direct proof and proof by contrapositive

A **direct** proof of $P \implies Q$ assumes P and reasons forward to Q . When that is awkward, Chapter 1 offers a free alternative: since an implication equals its contrapositive, proving $\neg Q \implies \neg P$ proves $P \implies Q$.

Method 1 (Proof by contrapositive). To prove $P \implies Q$: **assume** $\neg Q$, reason forward, and **derive** $\neg P$. Concluding $\neg Q \implies \neg P$ establishes the equivalent original $P \implies Q$.

Example — A contrapositive proof, written to template. Claim. If n^2 is even, then n is even.

Proof (contrapositive). Assume the negated conclusion: n is odd. Then $n = 2k + 1$ for some integer k , so

$$\begin{aligned} n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1, \end{aligned}$$

which is odd — the negated hypothesis. Thus “ n odd” implies “ n^2 odd”, and contraposition discharges the proof of “ n^2 even $\implies n$ even”. $\square$

4.3. Proof by exhaustion

Some statements split naturally into a handful of separate cases that together cover every possibility.

Proof by exhaustion (also called proof by cases) settles each case on its own; once the cases are exhausted, the statement holds in general.

Method 2 (Proof by exhaustion). To prove a statement S : split all possibilities into **finitely many cases** that are **exhaustive** (together they cover every possibility), prove S separately in each case, and conclude that S holds in all cases.

Example — A proof by cases on parity. Claim. For every integer n , the number $n^2 + n$ is even.

Proof (exhaustion on the parity of n). Every integer is even or odd — two exhaustive cases.

- If $n = 2k$ is even, then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$ is even.
- If $n = 2k + 1$ is odd, then $n^2 + n = (2k + 1)(2k + 2) = 2(2k + 1)(k + 1)$ is even.

Both cases yield an even number and no third case exists, so $n^2 + n$ is even for every integer n . $\square$

Pitfall. The cases must be **exhaustive**: miss one possibility and the proof is incomplete. Splitting an integer into “even” and “odd” covers every case; splitting into “ $n > 0$ ” and “ $n < 0$ ” silently forgets $n = 0$.

4.4. Proof by contradiction

Method 3 (Proof by contradiction). To prove a statement S : **assume** $\neg S$, derive a contradiction (a proposition that is both true and false, such as $r \neq r$), and conclude that $\neg S$ is impossible, so S holds.

Example — The irrationality of $\sqrt{2}$. **Claim.** $\sqrt{2}$ is irrational.

Proof (contradiction). Suppose, for contradiction, that $\sqrt{2} = p/q$ is rational, written in lowest terms (so p and q share no common factor). Then $p^2 = 2q^2$, so p^2 is even, hence p is even (by the example above), say $p = 2m$. Substituting, $4m^2 = 2q^2$, so $q^2 = 2m^2$ is even, hence q is even. But then p and q share the factor 2, contradicting “lowest terms”. The assumption is impossible, so $\sqrt{2}$ is irrational. $\square$

Pitfall. A proof by contradiction must reach a genuine **contradiction** — a proposition and its negation both forced true. Deriving something merely surprising, or re-deriving the assumption, proves nothing. State clearly which two facts collide.

4.5. Proof by induction

Induction proves a statement $P(n)$ for **every** whole number n at once. Picture an infinite line of dominoes: knock the first over (the base case), and guarantee that each falling domino topples the next (the inductive step); then all fall.

Method 4 (Proof by induction). To prove “ $P(n)$ holds for all $n \geq n_0$ ”:

- **Base case:** prove $P(n_0)$ directly.
- **Inductive step:** fix an arbitrary $n \geq n_0$, **assume** $P(n)$ (the **inductive hypothesis**), and prove $P(n + 1)$.

These two together license the conclusion “ $P(n)$ for all $n \geq n_0$ ”.

The inductive step does not assume what it is proving: it assumes $P(n)$ for a **single** generic n and deduces the **next** case $P(n + 1)$. That single implication, holding for every n , is what propagates the base case along the whole line.

Example — A first induction, written to template. Claim. For all $n \geq 1$, $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

Proof (induction on n). Base case ($n = 1$): the left side is 1 and the right side is $(1 \cdot 2)/2 = 1$. They agree.

Inductive step: fix $n \geq 1$ and assume $1 + \cdots + n = \frac{n(n+1)}{2}$. Adding $n + 1$ to both sides,

$$1 + \cdots + n + (n + 1) = \frac{n(n + 1)}{2} + (n + 1) = (n + 1)\left(\frac{n}{2} + 1\right) = \frac{(n + 1)(n + 2)}{2},$$

which is the formula with n replaced by $n + 1$.

By induction the formula holds for all $n \geq 1$. □

We now use induction for a result promised in Chapter 2.

Theorem 3 (The size of a power set — milestone proof). For every $n \in \mathbb{N}$ and every set X with $\text{Card}(X) = n$,

$$\text{Card}(\mathcal{P}(X)) = 2^n.$$

Proof. We argue by induction on $n = \text{Card}(X)$.

Base case ($n = 0$): the only set with no elements is $X = \emptyset$, whose sole subset is $\emptyset$ itself, so $\mathcal{P}(\emptyset) = \{\emptyset\}$ has $1 = 2^0$ element.

Inductive step: fix $n \geq 0$ and assume every n -element set has 2^n subsets. Let X have $n + 1$ elements. Pick one element $a \in X$ and set $Y = X \setminus \{a\}$, so $\text{Card}(Y) = n$. Every subset of X falls into exactly one of two cases:

- it does **not** contain a — then it is a subset of Y ; there are 2^n of these by the inductive hypothesis;
- it **does** contain a — then removing a leaves a subset of Y , and this pairing is a bijection between the subsets of X containing a and the subsets of Y ; so there are again 2^n of these.

The two cases are disjoint and exhaust $\mathcal{P}(X)$, so by the sum rule $\text{Card}(\mathcal{P}(X)) = 2^n + 2^n = 2^{n+1}$, which is the formula at $n + 1$.

By induction, $\text{Card}(\mathcal{P}(X)) = 2^n$ for all n . □

4.6. Analysis–synthesis

Many problems ask: **does a solution exist, and if so, which?** A two-phase method handles these.

Method 5 (Analysis–synthesis).

- **Analysis:** **assume** a solution exists and derive the conditions it must satisfy. This narrows the candidates — often to a single one — but proves nothing yet, because the assumption might describe nothing real.
- **Synthesis:** take each surviving candidate and **verify** it actually solves the problem. Only verified candidates are genuine solutions.

Analysis answers “**what could** a solution be?”; synthesis answers “**which of those really are?**”. Skipping synthesis is a real error, not a formality — analysis can produce phantom candidates that fail on substitution.

Example — Splitting a function into even and odd parts. Claim. Every function $f : \mathbb{R} \rightarrow \mathbb{R}$ can be written uniquely as $f = g + h$ with g even ($g(-x) = g(x)$) and h odd ($h(-x) = -h(x)$).

Analysis. Suppose such g, h exist. Then $f(x) = g(x) + h(x)$ and, replacing x by $-x$, $f(-x) = g(x) - h(x)$. Adding and subtracting these forces

$$g(x) = \frac{f(x) + f(-x)}{2}, \quad h(x) = \frac{f(x) - f(-x)}{2}.$$

So **if** a decomposition exists it must be this one — uniqueness is settled.

Synthesis. Define g, h by those formulas and check: $g(-x) = \frac{f(-x) + f(x)}{2} = g(x)$ so g is even; $h(-x) = \frac{f(-x) - f(x)}{2} = -h(x)$ so h is odd; and $g + h = f$. They work — existence is settled. The two phases together prove existence **and** uniqueness. $\square$

5. Sums and Products

5.1. A compact notation for long sums

Writing $1 + 2 + \dots + 100$ is clear enough; writing the sum of the first hundred squares that way is not. **Index notation** packs any such sum into a single symbol, and — more importantly — into a form we can manipulate with algebraic rules. This chapter turns those rules into fluency and proves the standard closed forms.

Definition 16 (Sum and product notation). For numbers $a_m, a_{m+1}, \dots, a_n$ (with $m \leq n$),

$$\sum_{k=m}^n a_k = a_m + a_{m+1} + \dots + a_n, \quad \prod_{k=m}^n a_k = a_m \times a_{m+1} \times \dots \times a_n.$$

The letter k is the **index**; it is a **dummy**, so $\sum_{k=1}^n a_k$ and $\sum_{i=1}^n a_i$ denote the same number. An **empty** sum (upper bound below lower) is 0, and an empty product is 1.

More generally, a sum may be indexed by any finite set I : the notation $\sum_{i \in I} a_i$ adds the a_i over all $i \in I$. When I is the **disjoint** union of two pieces J and K (so $J \cap K = \emptyset$), the sum breaks along the split,

$$\sum_{i \in J \cup K} a_i = \sum_{i \in J} a_i + \sum_{i \in K} a_i,$$

each term counted in exactly one piece. The bounded form $\sum_{k=m}^n$ is the special case $I = \llbracket m, n \rrbracket$.

5.2. Three rules for rearranging a sum

Proposition 5 (Linearity, reindexing, telescoping). For numbers a_k, b_k and constants λ, μ :

- **Linearity:**

$$\sum_{k=m}^n (\lambda a_k + \mu b_k) = \lambda \sum_{k=m}^n a_k + \mu \sum_{k=m}^n b_k.$$

- **Reindexing (shift):**

$$\sum_{k=m}^n a_k = \sum_{j=m+p}^{n+p} a_{j-p}$$

for any integer shift p — renaming the index changes the bounds, not the sum.

- **Telescoping:**

$$\sum_{k=m}^n (a_{k+1} - a_k) = a_{n+1} - a_m,$$

because every interior term cancels its neighbour.

Linearity lets you split a sum and pull out constants; it is the same distributive law as in Chapter 1, now over many terms. Reindexing is pure renaming — indispensable when two sums must be matched term by term. Telescoping is the most powerful of the three: it collapses an entire sum to two boundary terms.

Example — Telescoping in action. Compute $\sum_{k=1}^n \frac{1}{k(k+1)}$. The trick is **partial fractions**: $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$. With $a_k = 1/k$ this is $-(a_{k+1} - a_k)$, so

$$\begin{aligned} \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) &= -(a_{n+1} - a_1) \\ &= 1 - \frac{1}{n+1} \\ &= \frac{n}{n+1}. \end{aligned}$$

A sum with n terms collapses to a single fraction.

5.3. The same three moves for products

The product $\prod$ obeys the multiplicative echo of each rule above: replace each sum by a product, and **adding** a constant by **multiplying** by one.

Proposition 6 (Multiplicativity, reindexing, telescoping for products). For numbers a_k, b_k (nonzero wherever one sits under a division) and a constant c , taken over k from m to n — so there are $N = n - m + 1$ factors:

- **Multiplicativity:**

$$\prod_{k=m}^n (a_k b_k) = \left(\prod_{k=m}^n a_k \right) \left(\prod_{k=m}^n b_k \right), \quad \prod_{k=m}^n (c a_k) = c^N \prod_{k=m}^n a_k.$$

- **Reindexing (shift):**

$$\prod_{k=m}^n a_k = \prod_{j=m+p}^{n+p} a_{j-p}$$

for any integer shift p — renaming the index changes the bounds, not the product.

- **Telescoping:**

$$\prod_{k=m}^n \frac{a_{k+1}}{a_k} = \frac{a_{n+1}}{a_m},$$

because every interior factor cancels its neighbour.

A pulled-out constant comes out raised to the **number of factors** N , not as a plain multiple: that exponent is the product-world counterpart of the coefficient in linearity. Taking a logarithm makes the parallel exact — since $\ln \prod_{k=m}^n a_k = \sum_{k=m}^n \ln a_k$, every product identity is a sum identity read through $\ln$.

Example — A telescoping product. Compute $\prod_{k=1}^n \frac{k+1}{k}$. With $a_k = k$, each factor is a_{k+1}/a_k , so the product telescopes to the two boundary terms:

$$\prod_{k=1}^n \frac{k+1}{k} = \frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n} = \frac{n+1}{1} = n+1,$$

each numerator cancelling the next factor's denominator — the multiplicative twin of the telescoping sum above.

5.4. The standard sums

Telescoping also **derives** the closed forms for $\sum k$, $\sum k^2$, $\sum k^3$, without guessing. The idea: sum a telescoping identity whose interior produces the power you want.

Proposition 7 (The three standard sums). For all $n \geq 1$,

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

Proof. For the first, expand $(k+1)^2 - k^2 = 2k+1$ and sum from $k=1$ to n . The left side telescopes to $(n+1)^2 - 1 = n^2 + 2n$; the right side, by linearity, is $2\sum k + n$. Equating, $2\sum k + n = n^2 + 2n$, so $\sum k = (n^2 + n)/2 = n(n+1)/2$.

For the second, do the same with $(k+1)^3 - k^3 = 3k^2 + 3k + 1$. The left side telescopes to $(n+1)^3 - 1$; the right side is $3\sum k^2 + 3\sum k + n$. Substituting the known $\sum k$ and solving for $\sum k^2$ gives $n(n+1)(2n+1)/6$. The third is obtained identically from $(k+1)^4 - k^4$. $\square$

Remark. Notice $\sum k^3 = (\sum k)^2$: the sum of the first n cubes is the square of the n -th triangular number. It is a genuine identity, not a coincidence of small cases — the closed forms above prove it for every n .

5.5. The geometric sum

Theorem 4 (The geometric sum — milestone proof). For every $n \in \mathbb{N}$ and every real $q \neq 1$,

$$\sum_{k=0}^n q^k = \frac{1 - q^{n+1}}{1 - q}.$$

Proof. Let $S = \sum_{k=0}^n q^k = 1 + q + q^2 + \cdots + q^n$. Multiply by q and reindex:

$$qS = q + q^2 + \cdots + q^{n+1} = \sum_{k=1}^{n+1} q^k.$$

Subtract: in $S - qS$ every power from q^1 to q^n appears once with each sign and cancels, leaving only the two boundary terms

$$S - qS = q^0 - q^{n+1} = 1 - q^{n+1}.$$

Since $q \neq 1$ we may divide by $1 - q$:

$$S = \frac{1 - q^{n+1}}{1 - q}.$$

□

Pitfall. The formula needs $q \neq 1$; the division by $1 - q$ is illegal at $q = 1$. When $q = 1$ every term is 1, so the sum is simply $n + 1$. Always check the ratio before dividing.

5.6. The binomial theorem

The binomial coefficients of Chapter 3 earn their name here: they are precisely the coefficients in the expansion of a power of a sum.

Theorem 5 (The binomial theorem — milestone proof). For every $n \in \mathbb{N}$ and all real a, b ,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

Proof. We argue by induction on n , using Pascal's rule.

Base case ($n = 0$): the left side is $(a + b)^0 = 1$, and the right side is $\binom{0}{0} a^0 b^0 = 1$.

Inductive step: fix $n \geq 0$ and assume the formula for n . Then

$$(a + b)^{n+1} = (a + b) \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{k+1} b^{n-k} + \sum_{k=0}^n \binom{n}{k} a^k b^{n+1-k}.$$

Reindex the first sum with $j = k + 1$ (so $k = j - 1$, and j runs from 1 to $n + 1$):

$$\sum_{j=1}^{n+1} \binom{n}{j-1} a^j b^{n+1-j} + \sum_{j=0}^n \binom{n}{j} a^j b^{n+1-j}.$$

The term $j = n + 1$ (from the first sum) contributes $\binom{n}{n} a^{n+1} = a^{n+1}$; the term $j = 0$ (from the second) contributes $\binom{n}{0} b^{n+1} = b^{n+1}$; and for $j \in \llbracket 1, n \rrbracket$ both sums contribute, giving the coefficient $\binom{n}{j-1} + \binom{n}{j} = \binom{n+1}{j}$ by Pascal's rule. Collecting,

$$(a + b)^{n+1} = \sum_{j=0}^{n+1} \binom{n+1}{j} a^j b^{n+1-j},$$

which is the formula for $n + 1$.

By induction it holds for all $n \in \mathbb{N}$.

□

Example — Reading off an expansion. Row 4 of Pascal's triangle is 1, 4, 6, 4, 1, so

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$$

Setting $a = b = 1$ gives $\sum_{k=0}^4 \binom{4}{k} = 2^4 = 16$ — the total number of subsets of a 4-element set, recovering the power-set count of Chapter 4 as a special case.

6. Functions

6.1. A machine, then a definition

Think of a function as a machine: drop in an input, exactly one output comes out. The squaring machine sends $3 \mapsto 9$ and $-3 \mapsto 9$; it never sends 3 to two different numbers. That “exactly one output” is the whole idea.

Definition 17 (Function, domain, codomain, graph). A **function** is given by three data: a **domain** A , a **codomain** B , and a rule assigning to each $x \in A$ exactly one element $f(x) \in B$. We display it as

$$\begin{aligned} f : A &\longrightarrow B \\ x &\longmapsto f(x) \end{aligned}$$

The **range** (the image of the whole domain) is $\{f(x) : x \in A\} \subset B$, the outputs actually attained. The **graph** of f is the subset of $A \times B$

$$\Gamma_f = \{(x, f(x)) : x \in A\};$$

for $f : \mathbb{R} \longrightarrow \mathbb{R}$ this is the familiar curve in the plane.

Pitfall. Range and codomain are different. The function $f : \mathbb{R} \longrightarrow \mathbb{R}$, $x \mapsto x^2$, has codomain $\mathbb{R}$ but range $[0, +\infty)$ — negative numbers are never outputs. Many “is this surjective?” questions really ask whether the range fills the codomain.

6.2. Composition

To put on socks and then shoes is to perform one operation and feed its result to the next. **Composing** two functions chains them in exactly this way: the output of the first becomes the input of the second.

Definition 18 (Composition). Given $f : A \longrightarrow B$ and $g : B \longrightarrow C$, the **composite** $g \circ f : A \longrightarrow C$ is defined by

$$(g \circ f)(x) = g(f(x))$$

— do f first, then g . The order matters: $g \circ f$ and $f \circ g$ usually differ.

Example — Order matters. With $f : \mathbb{R} \longrightarrow \mathbb{R}$, $x \mapsto x + 1$, and $g : \mathbb{R} \longrightarrow \mathbb{R}$, $x \mapsto x^2$: $(g \circ f)(x) = (x + 1)^2$ but $(f \circ g)(x) = x^2 + 1$. Squaring-after-shifting is not shifting-after-squaring.

One composition never changes anything: composing with the function that returns its input untouched.

Definition 19 (Identity function). The **identity function** on a set A is

$$\begin{aligned} \text{id}_A : A &\longrightarrow A \\ x &\longmapsto x \end{aligned}$$

It leaves every element where it is. Composing with it has no effect: $f \circ \text{id}_A = f$ and $\text{id}_B \circ f = f$ for every $f : A \longrightarrow B$.

6.3. Image and preimage of a set

Definition 20 (Image and preimage). For $f : A \longrightarrow B$, a subset $S \subset A$, and a subset $T \subset B$:

- the **image** $f(S) = \{f(x) : x \in S\}$ — where S lands;
- the **preimage** $f^{-1}(T) = \{x \in A : f(x) \in T\}$ — everything that lands in T .

The preimage notation f^{-1} here needs **no** inverse function to exist; it is defined for any f .

Example — Image and preimage of squaring. For $f : \mathbb{R} \longrightarrow \mathbb{R}, x \mapsto x^2$: the image $f([1, 2]) = [1, 4]$. The preimage $f^{-1}([1, 4]) = [-2, -1] \cup [1, 2]$ — note it is **larger** than $[1, 2]$, because negatives also square into $[1, 4]$. Preimage and image are not inverse operations in general.

6.4. Injective, surjective, bijective

These three words classify functions by **how many inputs hit each output**.

Definition 21 (Injective, surjective, bijective). A function $f : A \longrightarrow B$ is

- **injective** (one-to-one) if different inputs give different outputs: $f(x_1) = f(x_2) \implies x_1 = x_2$;
- **surjective** (onto) if every $b \in B$ equals some $f(x)$ (the range is all of B);
- **bijective** if it is both — a perfect pairing of A with B .

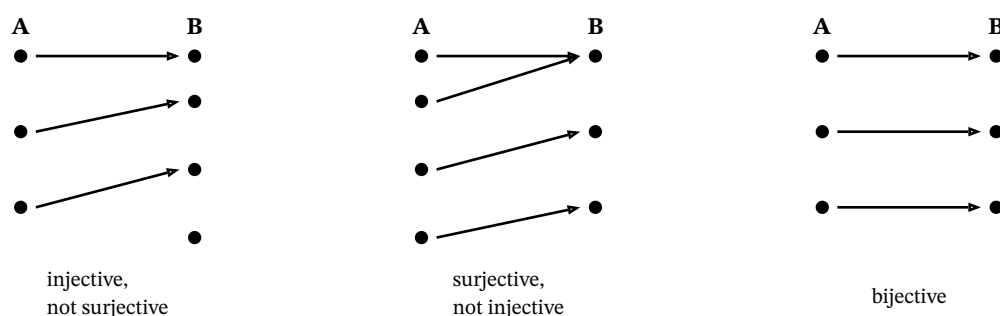


Figure 1: The three notions as arrow diagrams. Injective: no two arrows share a head. Surjective: every right-hand dot is hit. Bijective: a perfect matching, every dot on each side used exactly once.

Remark. Surjectivity depends on the **codomain** you declare. The map $x \mapsto x^2$ is not surjective as $\mathbb{R} \longrightarrow \mathbb{R}$, but **is** surjective as $\mathbb{R} \longrightarrow [0, +\infty)$. Shrinking the codomain to the range always makes a function surjective; restricting the domain can make it injective.

6.5. Inverse functions

Doubling a number and then halving the result returns the original number: halving **undoes** doubling. When a function pairs its two sets perfectly, there is always a companion function that runs it backwards this way — its **inverse**.

Definition 22 (Inverse function). If $f : A \longrightarrow B$ is **bijective**, its **inverse** $f^{-1} : B \longrightarrow A$ is the function that undoes it: $f^{-1}(y) = x$ exactly when $f(x) = y$. It satisfies $f^{-1}(f(x)) = x$ for all $x \in A$ and $f(f^{-1}(y)) = y$ for all $y \in B$.

Bijections also compose: if $f : A \longrightarrow B$ and $g : B \longrightarrow C$ are both bijective, then so is $g \circ f$, and its inverse undoes the two steps in reverse order,

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

The working criterion is worth stating on its own:

Theorem 6 (Characterisation of invertible functions). A function $f : A \longrightarrow B$ is bijective if and only if there is a function $g : B \longrightarrow A$ with $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$. When it exists, g is unique and equals f^{-1} .

Graphically, for $f : \mathbb{R} \longrightarrow \mathbb{R}$ the curve $y = f^{-1}(x)$ is the reflection of $y = f(x)$ across the line $y = x$ — input and output swap roles. To **compute** an inverse in practice, write $y = f(x)$, solve for x in terms of y , and read off $f^{-1}(y)$.

Example — Computing an inverse by solving. Let $f : \mathbb{R} \setminus \{3\} \longrightarrow \mathbb{R} \setminus \{2\}, x \longmapsto \frac{2x+1}{x-3}$. To invert it, fix a target value y and solve $y = \frac{2x+1}{x-3}$ for x . Because $x \neq 3$, each step is an equivalence:

$$\begin{aligned} y = \frac{2x+1}{x-3} &\iff y(x-3) = 2x+1 \\ &\iff yx - 3y = 2x+1 \\ &\iff x(y-2) = 3y+1 \\ &\iff x = \frac{3y+1}{y-2}. \end{aligned}$$

The last step divides by $y-2$, valid exactly when $y \neq 2$; and $y = 2$ is attained by no x , since it would force $0 = 7$. So the range of f is $\mathbb{R} \setminus \{2\}$ — which is why that set, not $\mathbb{R}$, is declared as the codomain: only then does the range fill the codomain, as a bijection requires (recall the pitfall above). Every $y \neq 2$ has the unique preimage just found, so the inverse is

$$\begin{aligned} f^{-1} : \mathbb{R} \setminus \{2\} &\longrightarrow \mathbb{R} \setminus \{3\} \\ y &\longmapsto \frac{3y+1}{y-2}. \end{aligned}$$

Check one value: $f(4) = 9/1 = 9$, and indeed $f^{-1}(9) = 28/7 = 4$.

6.6. The bijection theorem

For functions of a real variable there is a ready-made way to guarantee bijectivity — and hence an inverse — without solving any equation.

Theorem 7 (Bijection theorem). Let I be an interval of $\mathbb{R}$ and $f : I \longrightarrow \mathbb{R}$ be **continuous** and **strictly monotonic** (strictly increasing throughout, or strictly decreasing throughout). Then f is a bijection from I onto the interval $J = f(I)$, and its inverse $f^{-1} : J \longrightarrow I$ is continuous and strictly monotonic in the same direction.

Two words carry the hypotheses. **Continuous** means the graph has no jumps — you can draw it without lifting the pen; the notion is made precise in Section 9.2. **Strictly monotonic** means the graph only rises, or only falls. Under these two conditions injectivity is immediate — a strictly monotonic function never repeats a value — and continuity fills the range J with no gaps, so f maps I **onto** J . The complete justification of “no gaps” rests on a property of the real line taken up in a later analysis course; here we take the theorem as a working tool.

Pitfall. Both hypotheses are needed. Drop **continuity** and a monotonic function can skip values (a step function is strictly increasing on each piece yet hits nothing in the gaps). Drop **strict monotonicity** and injectivity fails: the map $x \longmapsto x^2$ on $\mathbb{R}$ takes the value 1 twice. Check both before invoking the theorem.

7. Comparing Infinities

7.1. Comparing sizes by pairing

Two finite sets have “the same number of elements” when their elements can be paired off, one to one, with none left over — that is, when a bijection runs between them. This pairing test needs no counting, so it keeps its meaning for **infinite** sets. Following it faithfully leads to a startling conclusion: some infinite sets are strictly larger than others.

7.2. Equipotence and countable sets

Definition 23 (Equipotence). Two sets A and B are **equipotent** when there exists a bijection $f : A \rightarrow B$. We then say they have the **same cardinality** and write $A \sim B$.

For finite sets, equipotence is just “the same number of elements”. The surprises begin with infinite sets, where a set can be equipotent to a **proper** subset of itself.

Example — The integers are equipotent to the naturals. Interleave the non-negative and negative integers. The map

$$\begin{aligned} f : \mathbb{N} &\longrightarrow \mathbb{Z} \\ n &\longmapsto \begin{cases} n/2 & \text{if } n \text{ is even} \\ -(n+1)/2 & \text{if } n \text{ is odd} \end{cases} \end{aligned}$$

sends $0, 1, 2, 3, 4, \dots \mapsto 0, -1, 1, -2, 2, \dots$, listing every integer exactly once. It is a bijection, so $\mathbb{Z} \sim \mathbb{N}$ — there are “as many” integers as naturals, though $\mathbb{N}$ sits strictly inside $\mathbb{Z}$.

Definition 24 (Countable set). A set is **countable** when it is finite or equipotent to $\mathbb{N}$. A set equipotent to $\mathbb{N}$ is **countably infinite**; concretely, its elements can be arranged in a single list $x_0, x_1, x_2, \dots$ indexed by $\mathbb{N}$, each element appearing exactly once.

The rationals, which look far denser than the naturals, are still countable: one can arrange the fractions in a grid and sweep it along successive diagonals, skipping repetitions, producing a single list of all of $\mathbb{Q}$. So $\mathbb{Q} \sim \mathbb{N}$ as well.

7.3. Cantor’s theorem

Is **every** set countable? No. No set — however large — can be paired with its own power set.

Theorem 8 (Cantor’s theorem). For every set X , there is no surjection from X onto its power set $\mathcal{P}(X)$. In particular X and $\mathcal{P}(X)$ are not equipotent: $\mathcal{P}(X)$ is strictly larger.

Proof. Let $f : X \rightarrow \mathcal{P}(X)$ be any function; we show f is not surjective by exhibiting a subset that is not a value of f . Consider the **diagonal set**

$$D = \{x \in X : x \notin f(x)\} \in \mathcal{P}(X).$$

Suppose, for contradiction, that D is a value of f , say $D = f(a)$ for some $a \in X$. Ask whether $a \in D$. By the definition of D ,

$$a \in D \iff a \notin f(a) = D,$$

so $a \in D$ holds exactly when $a \in D$ fails — a proposition equivalent to its own negation, which is impossible. Hence D is a value of no $f(a)$: the function f is not surjective. As f was arbitrary, no surjection $X \rightarrow \mathcal{P}(X)$ exists, and in particular no bijection. $\square$

Applied to $X = \mathbb{N}$, Cantor's theorem says $\mathcal{P}(\mathbb{N})$ is **not** countable. Since the real numbers can be shown equipotent to $\mathcal{P}(\mathbb{N})$, the set $\mathbb{R}$ is **uncountable**: its elements cannot be listed as $x_0, x_1, x_2, \dots$. There is, then, strictly more than one size of infinity — the countable infinity of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$, and the larger infinity of $\mathbb{R}$.

7.4. The Cantor–Bernstein theorem

Building an explicit bijection between two sets can be delicate. The following theorem replaces one hard bijection with two easier injections.

Theorem 9 (Cantor–Bernstein). Let A and B be sets. If there exist an injection $A \rightarrow B$ and an injection $B \rightarrow A$, then A and B are equipotent.

Its proof partitions A and B by tracing each element back and forth along the two injections and reassembling the pieces into a bijection; we admit it here and put it to use. The point is practical: to prove $A \sim B$ it suffices to embed each set into the other.

Example — An interval as large as the whole line. The open interval $(0, 1)$ is equipotent to $\mathbb{R}$. The inclusion $(0, 1) \rightarrow \mathbb{R}, x \mapsto x$, is an injection one way. For the other,

$$g : \mathbb{R} \rightarrow (0, 1), \quad x \mapsto \frac{1}{2} + \frac{1}{\pi} \arctan(x),$$

is injective (it is even strictly increasing). By Cantor–Bernstein, $(0, 1) \sim \mathbb{R}$: a bounded interval holds “as many” points as the entire real line.

8. Sequences

8.1. Real sequences

The list $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ never reaches 0, yet it settles toward 0 as tightly as we please. A **sequence** is such an infinite list of numbers, and the central question is whether it settles toward a single value — its **limit**. Making “settles toward” precise is the first genuine ε argument of the course, and the template for every limit that follows.

Definition 25 (Real sequence). A **real sequence** is a function

$$\begin{aligned} u : \mathbb{N} &\longrightarrow \mathbb{R} \\ n &\longmapsto u_n \end{aligned}$$

The number u_n is the **term of index** n , and the sequence itself is written $(u_n)_{n \in \mathbb{N}}$, or simply (u_n) . A sequence may be given **explicitly**, as $u_n = 1/(n+1)$, or **recursively**, by a first term u_0 and a rule $u_{n+1} = g(u_n)$ producing each term from the one before.

8.2. Convergence

Definition 26 (Convergence of a sequence). A sequence (u_n) **converges** to a real number ℓ when

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, |u_n - \ell| < \varepsilon.$$

We then call ℓ the **limit** and write $\lim_{n \rightarrow \infty} u_n = \ell$, or $u_n \rightarrow \ell$. A sequence that converges to some real number is **convergent**; otherwise it **diverges**.

Read the definition as a challenge and a response. An adversary names a tolerance $\varepsilon > 0$, however small; you must supply a rank N beyond which **every** term lies within ε of ℓ . The smaller the ε , the larger the N you may need — but one must always exist.

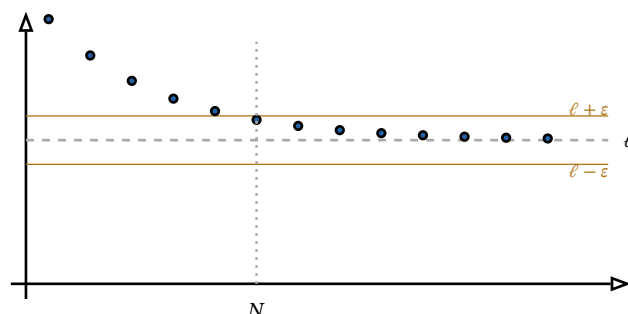


Figure 2: Convergence to ℓ . For the tolerance ε shown, all terms from the rank N onward lie inside the band $(\ell - \varepsilon, \ell + \varepsilon)$. A smaller ε narrows the band and pushes N to the right.

A limit, once it exists, is unique.

Theorem 10 (Uniqueness of the limit). A convergent sequence has exactly one limit.

Proof. Suppose $u_n \rightarrow \ell$ and $u_n \rightarrow \ell'$, and let $\varepsilon > 0$. There is a rank N_1 beyond which $|u_n - \ell| < \varepsilon$, and a rank N_2 beyond which $|u_n - \ell'| < \varepsilon$. For any $n \geq \max(N_1, N_2)$, the triangle inequality gives

$$|\ell - \ell'| \leq |\ell - u_n| + |u_n - \ell'| < 2\varepsilon.$$

Thus $|\ell - \ell'|$ is smaller than every positive number 2ε , which forces $|\ell - \ell'| = 0$, that is $\ell = \ell'$. $\square$

Theorem 11 (A convergent sequence is bounded). If (u_n) converges, then it is bounded: there is a real M with $|u_n| \leq M$ for all n .

Proof. Let $\ell = \lim_{n \rightarrow \infty} u_n$. Apply the definition with the particular tolerance $\varepsilon = 1$: there is a rank N with $|u_n - \ell| < 1$ for all $n \geq N$, hence $|u_n| < |\ell| + 1$ for those n . Only the finitely many terms $u_0, \dots, u_{N-1}$ remain, so

$$M = \max(|u_0|, |u_1|, \dots, |u_{N-1}|, |\ell| + 1)$$

is a bound satisfied by every term. The decisive step is choosing a single concrete ε , which corrals all but finitely many terms into one band. $\square$

The converse fails: the bounded sequence $u_n = (-1)^n$ does not converge; its terms are alternately -1 and 1 , so they approach no single value.

8.3. Operations on limits

Limits combine with arithmetic exactly as one would hope.

Proposition 8 (Operations on limits). If $u_n \rightarrow \ell$ and $v_n \rightarrow \ell'$, then as $n \rightarrow \infty$:

$$u_n + v_n \rightarrow \ell + \ell', \quad \lambda u_n \rightarrow \lambda \ell \quad (\lambda \in \mathbb{R}), \quad u_n v_n \rightarrow \ell \ell',$$

and, provided $\ell' \neq 0$ and the terms are defined, $u_n/v_n \rightarrow \ell/\ell'$.

Proof. We prove the sum; the others are similar. Let $\varepsilon > 0$. Since $u_n \rightarrow \ell$, there is N_1 with $|u_n - \ell| < \varepsilon/2$ for $n \geq N_1$; since $v_n \rightarrow \ell'$, there is N_2 with $|v_n - \ell'| < \varepsilon/2$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$,

$$|(u_n + v_n) - (\ell + \ell')| \leq |u_n - \ell| + |v_n - \ell'| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Splitting the tolerance in half for each sequence is the standard device. $\square$

These rules cover a limit only when every ingredient converges to a finite value. When a limit involves ∞ , four combinations escape the rules and must be studied case by case.

Pitfall. The four **indeterminate forms**

$$\infty - \infty, \quad 0 \times \infty, \quad \frac{\infty}{\infty}, \quad \frac{0}{0}$$

have no automatic value: the answer depends on the specific sequences. For instance $\sqrt{n^2 + n} - n$ is of the form $\infty - \infty$, yet

$$\sqrt{n^2 + n} - n = \frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{\sqrt{1 + 1/n} + 1} \rightarrow \frac{1}{2},$$

so it converges to $1/2$ — neither 0 nor ∞ . Never assign an indeterminate form a value without resolving it.

8.4. The squeeze theorem and monotone convergence

Two theorems let us settle convergence without exhibiting the limit's value directly.

Theorem 12 (Squeeze theorem). Let (u_n) , (a_n) , (b_n) be sequences with $a_n \leq u_n \leq b_n$ for all n large enough. If $a_n \rightarrow \ell$ and $b_n \rightarrow \ell$ (the **same** limit), then $u_n \rightarrow \ell$.

Proof. Let $\varepsilon > 0$. Beyond some rank, $|a_n - \ell| < \varepsilon$ and $|b_n - \ell| < \varepsilon$, so $\ell - \varepsilon < a_n$ and $b_n < \ell + \varepsilon$. With $a_n \leq u_n \leq b_n$ this gives $\ell - \varepsilon < u_n < \ell + \varepsilon$, that is $|u_n - \ell| < \varepsilon$. Hence $u_n \rightarrow \ell$. $\square$

Example — Squeezing to a limit. For $u_n = \frac{\sin n}{n}$, the bound $|\sin n| \leq 1$ gives $-1/n \leq u_n \leq 1/n$. Both $-1/n$ and $1/n$ tend to 0 , so by the squeeze theorem $u_n \rightarrow 0$, even though $\sin n$ itself has no limit.

Definition 27 (Monotone sequence). A sequence (u_n) is **increasing** if $u_n \leq u_{n+1}$ for all n , and **decreasing** if $u_n \geq u_{n+1}$ for all n . It is **monotone** if it is one or the other.

Theorem 13 (Monotone convergence). A monotone bounded sequence converges. More precisely, an increasing sequence bounded above converges to the least of its upper bounds; a decreasing sequence bounded below converges to the greatest of its lower bounds. An increasing sequence that is **not** bounded above tends to $+\infty$.

The existence of that least upper bound is the deep property of the real line — its **completeness** — established in a later analysis course; here we take the theorem as a working tool. It is the standard way to prove a recursively defined sequence converges, since often one can show it monotone and bounded without ever computing its terms.

8.5. Arithmetico-geometric sequences

One recursive family is fully solvable in closed form.

Definition 28 (Arithmetico-geometric sequence). An **arithmetico-geometric sequence** is one defined by $u_0 \in \mathbb{R}$ and the recurrence $u_{n+1} = au_n + b$, with $a, b \in \mathbb{R}$. When $a = 1$ it is arithmetic; when $b = 0$ it is geometric.

Assume $a \neq 1$. The method has three steps: find the **fixed point**, subtract it to expose a geometric sequence, then read off the limit.

Proposition 9 (Closed form and convergence). Let $u_{n+1} = au_n + b$ with $a \neq 1$, and let $\ell = b/(1 - a)$ be the unique solution of $\ell = a\ell + b$ (the **fixed point**). Then

$$u_n = a^n(u_0 - \ell) + \ell \quad \text{for all } n \in \mathbb{N},$$

and (u_n) converges if and only if $|a| < 1$ or $u_0 = \ell$; when $|a| < 1$ the limit is ℓ .

Proof. Set $v_n = u_n - \ell$. Using $\ell = a\ell + b$,

$$v_{n+1} = u_{n+1} - \ell = au_n + b - (a\ell + b) = a(u_n - \ell) = av_n,$$

so (v_n) is geometric with ratio a , giving $v_n = a^n v_0$. Therefore $u_n = \ell + a^n(u_0 - \ell)$. If $|a| < 1$ then $a^n \rightarrow 0$, so $u_n \rightarrow \ell$; if $u_0 = \ell$ the sequence is constant equal to ℓ . Otherwise $|a| \geq 1$ and $u_0 \neq \ell$, so $a^n(u_0 - \ell)$ does not tend to a finite limit, and (u_n) diverges. $\square$

Example — Solving a recurrence. Let $u_0 = 0$ and $u_{n+1} = \frac{1}{2}u_n + 3$. The fixed point solves $\ell = \frac{1}{2}\ell + 3$, so $\ell = 6$. Then $u_n = 6 + (1/2)^n(0 - 6) = 6(1 - 2^{-n})$. Since $|1/2| < 1$, the sequence converges to 6; and indeed $u_1 = 3$, $u_2 = 4.5$, $u_3 = 5.25$ climb toward 6.

9. Differentiation

9.1. Working near a point

Differentiation and continuity both describe a function's behaviour close to a point, so we fix that phrase first.

Definition 29 (Near a point). A property holds **near a point** a (equivalently, **for x near a**) when it holds for every x in some open interval $(a - \delta, a + \delta)$ around a , for some $\delta > 0$. A function is **defined near a** when such an interval lies in its domain.

9.2. Limits and continuity

Definition 30 (Limit of a function). Let f be defined near a (except possibly at a). We say $f(x)$ **tends to** $\ell \in \mathbb{R}$ **as x tends to a** , written $\lim_{x \rightarrow a} f(x) = \ell$, when

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x, 0 < |x - a| < \delta \implies |f(x) - \ell| < \varepsilon.$$

At the bounds of the domain the variants read the same way. Tending to $+\infty$: replace “ $0 < |x - a| < \delta$ ” by “ $x > A$ ” for a threshold A . Growing without bound: replace “ $|f(x) - \ell| < \varepsilon$ ” by “ $f(x) > B$ ” for a threshold B . Thus $\lim_{x \rightarrow +\infty} 1/x = 0$ and $\lim_{x \rightarrow 0^+} 1/x = +\infty$.

Definition 31 (Continuity at a point). Let f be defined near a , with a in its domain. Then f is **continuous at a** when

$$\lim_{x \rightarrow a} f(x) = f(a).$$

It is **continuous on an interval I** when it is continuous at every point of I . Intuitively, the graph makes no jump at a : near inputs give near outputs.

Continuity has an equivalent phrasing through sequences, which is often the easiest to use — and which links this chapter to Chapter 8.

Theorem 14 (Sequential characterisation of continuity). Let f be defined near a . Then f is continuous at a if and only if, for **every** sequence (x_n) of points of the domain with $x_n \rightarrow a$, one has $f(x_n) \rightarrow f(a)$.

Proof. Assume f continuous at a and take $x_n \rightarrow a$. Let $\varepsilon > 0$. Continuity gives a $\delta > 0$ with $|f(x) - f(a)| < \varepsilon$ whenever $|x - a| < \delta$; and $x_n \rightarrow a$ gives a rank N beyond which $|x_n - a| < \delta$. For $n \geq N$ we get $|f(x_n) - f(a)| < \varepsilon$, so $f(x_n) \rightarrow f(a)$.

Conversely, we prove the contrapositive: if f is **not** continuous at a , some $\varepsilon > 0$ resists every δ , so for each n the choice $\delta = 1/(n + 1)$ yields a point x_n with $|x_n - a| < 1/(n + 1)$ yet $|f(x_n) - f(a)| \geq$

ε . Then $x_n \rightarrow a$ while $f(x_n)$ does not tend to $f(a)$, breaking the sequential condition. Contraposition discharges the proof. $\square$

Because limits pass through sums, products, and quotients (Chapter 8), continuity does too.

Proposition 10 (Operations preserving continuity). If f and g are continuous at a , then so are $f + g$, λf (for $\lambda \in \mathbb{R}$), and fg ; and f/g is continuous at a when $g(a) \neq 0$. If f is continuous at a and g is continuous at $f(a)$, then $g \circ f$ is continuous at a . Consequently every polynomial is continuous on $\mathbb{R}$, and every rational function is continuous wherever its denominator does not vanish.

9.3. The derivative

How fast is f changing at $x = a$? Average the change over a small step h and shrink the step. The average rate $\frac{f(a+h)-f(a)}{h}$ is the slope of the **secant** line through two nearby points on the graph; as $h \rightarrow 0$ the secant pivots into the **tangent**, and its slope is the derivative.

Definition 32 (Difference quotient and derivative). For f defined near a , the **difference quotient** at a is

$$\frac{f(a+h)-f(a)}{h}, \quad h \neq 0.$$

If it tends to a finite limit as $h \rightarrow 0$, then f is **differentiable at a** and that limit is the **derivative**

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h},$$

the slope of the **tangent line** to the graph at $(a, f(a))$.

The tangent line is $y = f(a) + f'(a)(x - a)$: the line through $(a, f(a))$ with slope $f'(a)$. What is it **for**? Rearranged, it says

$$f(a+h) \approx f(a) + f'(a)h \quad \text{for small } h,$$

so the tangent is the **linear approximation** of f near a — the straight line that best matches the curve there, and the basis of every “small change” estimate. Later in this chapter we shall see a second use: when f is convex, its graph never dips below any tangent, which turns the tangent into a supply of inequalities.

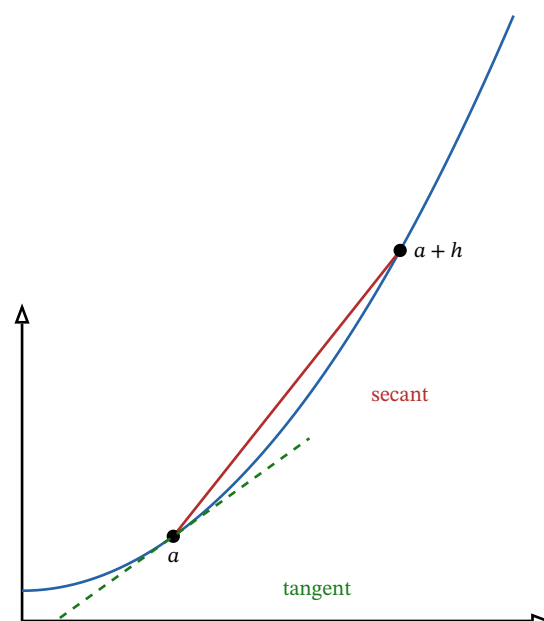


Figure 3: As $h \rightarrow 0$ the secant (red) through $(a, f(a))$ and $(a + h, f(a + h))$ rotates toward the tangent (green). The derivative $f'(a)$ is the tangent's slope.

9.4. Differentiability and continuity

Theorem 15 (Differentiable implies continuous). If f is differentiable at a , then f is continuous at a . The converse fails.

Proof. As $h \rightarrow 0$,

$$f(a + h) - f(a) = \frac{f(a + h) - f(a)}{h} \cdot h \rightarrow f'(a) \cdot 0 = 0,$$

so $f(a + h) \rightarrow f(a)$, which is continuity at a . $\square$

Continuity is strictly weaker: a continuous graph may still fail to have a tangent, in three recognisable ways.

Pitfall. Continuity does **not** imply differentiability. Three standard failures at a point where f is nonetheless continuous:

- an **angular point** (corner): the map $x \mapsto |x|$ fails at 0, where the difference quotient equals -1 for $h < 0$ and $+1$ for $h > 0$, so it has no single limit;
- a **vertical tangent**: the map $x \mapsto \sqrt[3]{x}$ fails at 0, where the quotient $h^{1/3}/h = h^{-2/3} \rightarrow +\infty$, so no finite slope exists;
- an **oscillating quotient**: the map $x \mapsto x \sin(1/x)$, extended by the value 0 at 0, fails at 0, where the quotient is $\sin(1/h)$, which oscillates in $[-1, 1]$ without settling.

9.5. The differentiation rules

The derivatives of the basic functions are known from high school; the rules below combine them, so that any function assembled from the basics can be differentiated mechanically.

Proposition 11 (Rules of differentiation). For differentiable f, g and a constant c :

$$\begin{aligned}(f + g)' &= f' + g', & (cf)' &= cf', \\ (fg)' &= f'g + fg' & & \text{(product rule),} \\ \left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} & & \text{(quotient rule, } g \neq 0), \\ (g \circ f)' &= (g' \circ f) \cdot f' & & \text{(chain rule).}\end{aligned}$$

The basic library: $(x^n)' = nx^{n-1}$, $(e^x)' = e^x$, $(\ln x)' = 1/x$, $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$.

Pitfall. The product rule is **not** $f'g'$, and the quotient rule's numerator order matters — it is $f'g - fg'$, not $fg' - f'g$. A sign error here spreads through an entire problem. The chain rule's commonest slip is forgetting the inner factor f' : the derivative of $\sin(2x)$ is $2\cos(2x)$, not $\cos(2x)$.

Example — Differentiating a quotient. Differentiate $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \frac{x}{x^2+1}$. Take numerator $u = x$ (so $u' = 1$) and denominator $v = x^2 + 1$ (so $v' = 2x$); the quotient rule gives

$$f'(x) = \frac{u'v - uv'}{v^2} = \frac{1 \cdot (x^2 + 1) - x \cdot 2x}{(x^2 + 1)^2} = \frac{1 - x^2}{(x^2 + 1)^2}.$$

Since $(x^2 + 1)^2 > 0$, the sign of f' is the sign of $1 - x^2$: positive on $(-1, 1)$, negative outside it.

Example — The chain rule on a composition. To differentiate $x \mapsto (x^2 + 1)^5$: outer $g(u) = u^5$, inner $f(x) = x^2 + 1$, giving $5(x^2 + 1)^4 \cdot 2x = 10x(x^2 + 1)^4$. For $x \mapsto e^{3x}$, the inner is $3x$ with derivative 3, so the derivative is $3e^{3x}$.

When a derivative is itself continuous, the function earns a name we shall need for integration.

Definition 33 (Class C^1). A function $f : I \rightarrow \mathbb{R}$ is **of class C^1** on the interval I when f is differentiable on I and its derivative f' is continuous on I . One also says f is **continuously differentiable**.

9.6. Monotonicity from the sign of the derivative

The derivative reveals where a function rises and falls.

Theorem 16 (Sign of the derivative and monotonicity). Let f be differentiable on an interval I .

- If $f' \geq 0$ throughout I , then f is increasing on I ; if $f' > 0$ throughout, f is **strictly** increasing.

- If $f' \leq 0$ throughout I , then f is decreasing; if $f' < 0$ throughout, strictly decreasing.
- If $f' = 0$ throughout, then f is constant.

Geometrically this is immediate: a positive slope everywhere means the graph only climbs. The rigorous derivation uses the Mean Value Theorem, a result of the next analysis course; here we state the link and put it to work. Combined with the bijection theorem of Chapter 6, “ $f' > 0$ on an interval” certifies that f is a bijection onto its image — this is how $\exp : \mathbb{R} \rightarrow (0, +\infty)$, with derivative $e^x > 0$ everywhere, is seen to be a bijection with inverse the natural logarithm.

All of a function’s rise-and-fall behaviour fits on one grid called a **variation table**. Its top row marks the ends of the interval and the points where f' vanishes, in increasing order; its middle row gives the sign of f' on each subinterval; its bottom row turns each sign into an arrow — $\nearrow$ where $f' > 0$ so f rises, $\searrow$ where $f' < 0$ so f falls — and writes the value of f at each marked point.

x	$-\infty$		-1		1		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
f	$-\infty$	$\nearrow$	2	$\searrow$	-2	$\nearrow$	$+\infty$

Table 3: The variation table of $x \mapsto x^3 - 3x$. Top row: the marked points. Middle row: the sign of $f'(x) = 3(x-1)(x+1)$ on each subinterval. Bottom row: the resulting motion of f , with its value at each mark.

Example — Reading a variation table. The table above is built for $f : \mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x^3 - 3x$, whose derivative $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$ is positive on $(-\infty, -1)$, negative on $(-1, 1)$, and positive on $(1, +\infty)$, vanishing at $x = \pm 1$. Reading the bottom row: f climbs to $f(-1) = 2$, falls to $f(1) = -2$, then climbs again — a crest at $x = -1$ and a trough at $x = 1$.

9.7. Local extrema and Fermat’s condition

Definition 34 (Local extremum). A function f has a **local maximum** at c if $f(x) \leq f(c)$ for all x near c , and a **local minimum** if $f(x) \geq f(c)$ for all x near c . A **local extremum** is either.

At a crest or trough in the interior of the domain, the tangent is horizontal. This is Fermat’s condition — a necessary condition that locates the candidates for an extremum.

Theorem 17 (Fermat’s condition — milestone proof). Let I be an interval of $\mathbb{R}$, $f : I \rightarrow \mathbb{R}$, and $a \in I$ **interior** to I (not an endpoint). If f is differentiable at a and has a local extremum at a , then $f'(a) = 0$.

Proof. Suppose f has a local **maximum** at a (the minimum case is identical with inequalities reversed). Then $f(a+h) - f(a) \leq 0$ for all h small enough, and since a is interior, both signs of h are allowed.

- For small $h > 0$:

$$\frac{f(a+h) - f(a)}{h} \leq 0,$$

so letting $h \rightarrow 0^+$ gives $f'(a) \leq 0$.

- For small $h < 0$: dividing the same numerator ≤ 0 by the negative h reverses the inequality,

$$\frac{f(a+h) - f(a)}{h} \geq 0,$$

so letting $h \rightarrow 0^-$ gives $f'(a) \geq 0$.

Since f is differentiable at a , both one-sided limits equal the single number $f'(a)$. From $f'(a) \leq 0$ and $f'(a) \geq 0$ we conclude $f'(a) = 0$. $\square$

Pitfall. Fermat's condition is **necessary, not sufficient**: $f'(c) = 0$ does not force an extremum. For $x \mapsto x^3$, the derivative at 0 is zero, yet 0 is neither a maximum nor a minimum — the curve merely flattens and continues rising. And the condition concerns **interior** points: an extremum can also sit at an **endpoint** of the domain, where f' need not vanish. To find extrema, examine the points where $f' = 0$ **and** the endpoints.

Example — Locating and confirming extrema. Find the local extrema of $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x^3 - 3x$. By Fermat, they can only occur where $f'(x) = 3(x-1)(x+1) = 0$, i.e. at $x = -1$ and $x = 1$. The sign of f' (read from Table 3) changes from $+$ to $-$ at $x = -1$: a local **maximum**, value $f(-1) = 2$. It changes from $-$ to $+$ at $x = 1$: a local **minimum**, value $f(1) = -2$. The sign change confirms each candidate is a genuine extremum.

9.8. The second derivative: convexity and concavity

Differentiating f' in turn measures how the **slope** itself changes.

Definition 35 (Second derivative). If f' is itself differentiable on I , its derivative $(f')'$ is the **second derivative** of f , written f'' .

Convexity is about the shape of the graph — whether it bends upward like a bowl — and is defined through chords, not through any derivative.

Definition 36 (Convex and concave function). A function $f : I \rightarrow \mathbb{R}$ is **convex** on the interval I when, for all $x, y \in I$ and all $t \in [0, 1]$,

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) :$$

every chord of the graph lies on or above the graph. It is **concave** when the reverse inequality holds, so that every chord lies on or below the graph.

For a twice-differentiable function the second derivative decides convexity, and the tangent gives a matching family of inequalities.

Proposition 12 (Convexity through the second derivative). Let f be twice differentiable on an interval I .

- f is convex on I if and only if $f'' \geq 0$ throughout I ; it is concave if and only if $f'' \leq 0$ throughout I .
- If f is convex on I , then for every $a \in I$ its graph lies on or above the tangent at a :

$$f(x) \geq f(a) + f'(a)(x - a) \quad \text{for all } x \in I.$$

We take these from the Mean Value Theorem of the next analysis course; the second statement is the **convexity inequality** announced earlier — a convex graph never dips below a tangent, so each tangent yields a lower bound for f .

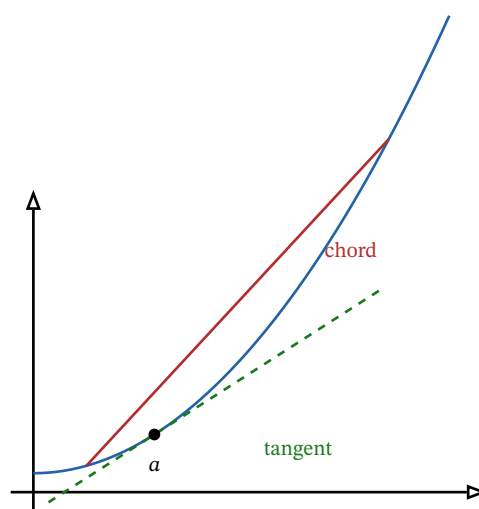


Figure 4: A convex function. Every chord (red) lies above the graph; every tangent (green) lies below it. The gap between graph and tangent is what the convexity inequality bounds.

Definition 37 (Inflection point). A point a interior to I is an **inflection point** of f when the convexity of f changes there — from convex to concave, or the reverse. For a twice-differentiable f , this is where f'' changes sign.

Example — Convexity of a cubic. For $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x^3 - 3x$, one has $f''(x) = 6x$. So $f'' < 0$ on $(-\infty, 0)$ — concave — and $f'' > 0$ on $(0, +\infty)$ — convex — with an inflection point at $x = 0$, where the graph crosses from bending down to bending up.

10. Integration

10.1. Signed area and antiderivatives

Differentiation measures **rate**; integration measures **accumulation** — total distance from a speed, total area under a curve. The definite integral $\int_a^b f(x) dx$ is the **signed area** between the graph of f and the horizontal axis from a to b : area above the axis counts positive, area below counts negative. The climax of elementary calculus is that this accumulation is undone by differentiation.

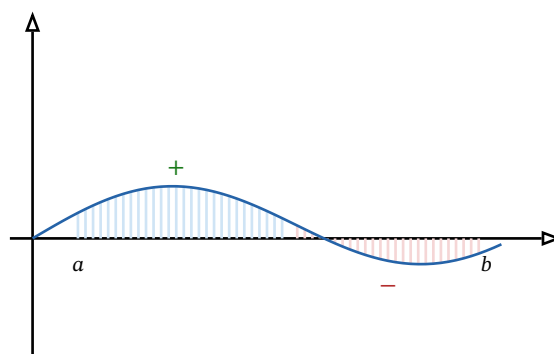


Figure 5: The definite integral as signed area. Where the graph is above the axis (blue-shaded) the area counts positively; where it dips below (red-shaded) it counts negatively. $\int_a^b f$ is the positive area minus the negative area.

Remark. The variable under the integral sign is a **dummy**, exactly like the index of a sum: it is bound by the integral and invisible from outside. Thus

$$\int_a^b f(t) dt = \int_a^b f(x) dx = \int_a^b f(u) du,$$

all one and the same number. The letter is a private name for the running point, free to be renamed.

Definition 38 (Antiderivative and indefinite integral). An **antiderivative** of f on an interval is a function F with $F' = f$. Any two antiderivatives of f differ by a constant, so we write the **indefinite integral**

$$\int f(x) dx = F(x) + C,$$

the whole family of antiderivatives, with C an arbitrary constant.

10.2. Recognising usual forms

Reading the differentiation library backwards gives a short list of antiderivatives to recognise on sight; the full tables close the book.

Framework 1 (A short table of antiderivatives). Up to an added constant C :

$$\int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1), \quad \int \frac{1}{x} dx = \ln|x|,$$

$$\int e^x dx = e^x, \quad \int \cos x dx = \sin x, \quad \int \sin x dx = -\cos x.$$

Example — Recognising a form.

$$\int (3x^2 - 4x + 5) dx = x^3 - 2x^2 + 5x + C,$$

term by term via the power rule. Differentiating the answer returns the integrand — the surest check.

10.3. The Fundamental Theorem of Calculus

Theorem 18 (Fundamental Theorem of Calculus). Let f be continuous on $[a, b]$.

1. The **accumulation function** $G(x) = \int_a^x f(t) dt$ is an antiderivative of f : $G'(x) = f(x)$.
2. (**Evaluation**) If F is **any** antiderivative of f , then

$$\int_a^b f(x) dx = F(b) - F(a) =: [F(x)]_a^b.$$

Part 1 says differentiation and integration are inverse processes; Part 2 turns the signed area — a limit of thin rectangles — into simple arithmetic: find an antiderivative and subtract its values at the endpoints.

Example — Area under a parabola. For $\int_0^1 x^2 dx$, an antiderivative of x^2 is $x^3/3$, so the area is $[x^3/3]_0^1 = 1/3 - 0 = 1/3$. No rectangles required — the theorem did the work.

Pitfall. Part 2 requires f to be **continuous** on the whole segment $[a, b]$. Applying it across a point where f is **unbounded** — as in $\int_{-1}^1 (1/x^2) dx$, where $1/x^2 \rightarrow +\infty$ as $x \rightarrow 0$ — produces a meaningless answer: the hypothesis fails, so the conclusion does not hold. Check continuity on the entire segment before evaluating.

10.4. Linearity, Chasles, and the bounds

Three structural properties let us cut and reassemble integrals.

Proposition 13 (Linearity, Chasles' relation, exchange of bounds). For f, g continuous and constants λ, μ :

- **Linearity:**

$$\int_a^b (\lambda f + \mu g) = \lambda \int_a^b f + \mu \int_a^b g.$$

- **Chasles' relation:** for any c ,

$$\int_a^b f + \int_b^c f = \int_a^c f.$$

- **Exchange of bounds:**

$$\int_b^a f = - \int_a^b f, \quad \text{and in particular} \quad \int_a^a f = 0.$$

The exchange rule keeps Chasles' relation consistent for every arrangement of a, b, c on the line, whether or not b lies between a and c .

Example — Splitting at a corner with Chasles. Compute $\int_0^2 |x - 1| dx$. The integrand changes formula at $x = 1$: it is $1 - x$ on $[0, 1]$ and $x - 1$ on $[1, 2]$. Chasles' relation splits the integral there:

$$\int_0^2 |x - 1| dx = \int_0^1 (1 - x) dx + \int_1^2 (x - 1) dx = \left[x - \frac{x^2}{2} \right]_0^1 + \left[\frac{x^2}{2} - x \right]_1^2 = \frac{1}{2} + \frac{1}{2} = 1.$$

Example — Reversing the bounds. Reversing the limits negates the integral: since $\int_0^2 x dx = [x^2/2]_0^2 = 2$, the exchange rule gives $\int_2^0 x dx = -2$, and $\int_2^2 x dx = 0$.

10.5. Integration by parts

Reversing the product rule gives one of the two central techniques.

Theorem 19 (Integration by parts — milestone proof). Let I be an interval, $u, v : I \rightarrow \mathbb{R}$ of class C^1 , and $a, b \in I$. Then

$$\int_a^b u'(t)v(t) dt = [u(t)v(t)]_a^b - \int_a^b u(t)v'(t) dt.$$

Proof. By the product rule, $(uv)' = u'v + uv'$ on I . Since u, v are of class C^1 , both $u'v$ and uv' are continuous, so uv is an antiderivative of the continuous function $u'v + uv'$. By the Fundamental Theorem (Part 2),

$$\int_a^b (u'(t)v(t) + u(t)v'(t)) dt = [u(t)v(t)]_a^b.$$

Splitting the left side by linearity and moving $\int_a^b uv'$ to the right gives the stated identity. $\square$

The art is choosing which factor plays v (to be differentiated) and which plays u' (to be integrated): pick v so that v' is **simpler**.

Example — Integrating xe^x . Take $v = x$ (so $v' = 1$, simpler) and $u' = e^x$ (so $u = e^x$). Then

$$\int xe^x dx = xe^x - \int 1 \cdot e^x dx = xe^x - e^x + C = (x - 1)e^x + C.$$

A standard trick uses the hidden factor 1: for $\int \ln x dx$, take $v = \ln x$, $u' = 1$, giving $x \ln x - \int x \cdot (1/x) dx = x \ln x - x + C$.

10.6. Substitution

Reversing the chain rule gives **substitution** (the change of variable).

Theorem 20 (Integration by substitution). If $u = \varphi(x)$ with φ of class C^1 , then $du = \varphi'(x) dx$ and

$$\int f(\varphi(x))\varphi'(x) dx = \int f(u) du.$$

For a definite integral, change the bounds too: x from a to b becomes u from $\varphi(a)$ to $\varphi(b)$.

Example — Spotting the inner function. For $\int 2x \cos(x^2) dx$, put $u = x^2$, so $du = 2x dx$, which is exactly present. The integral becomes $\int \cos u du = \sin u + C = \sin(x^2) + C$. Substitution works because the integrand already contains the derivative of the inner function.

Example — Changing the bounds. For $\int_1^2 2xe^{x^2} dx$, put $u = x^2$, so $du = 2x dx$ and the bounds move with the variable: $x = 1$ gives $u = 1$, and $x = 2$ gives $u = 4$. Hence

$$\int_1^2 2xe^{x^2} dx = \int_1^4 e^u du = [e^u]_1^4 = e^4 - e.$$

Pitfall. In a **definite** integral, either convert the bounds to the new variable (cleanest) **or** return to x before substituting the original bounds — but never feed the old x -bounds into a u -expression. Mixing the two is a frequent and costly error.

Notation reference

A compact glossary of the symbols used in this book, with the chapter where each is introduced. Read the middle column aloud: that is how the symbol is spoken.

Symbol	Read as	Ch.
$\neg, \wedge, \vee$	not; and; or	1
$\implies, \iff$	implies; if and only if	1
$\forall, \exists, \exists!$	for all; there exists; there exists a unique	1
$\in, \notin$	is / is not an element of	2
$\subset$	is a subset of	2
$\cup, \cap$	union; intersection	2
$\setminus, A^c$	set difference; complement of A	2
$\emptyset$	the empty set	2
$\{x \in A : P(x)\}$	the x in A such that $P(x)$ holds	2
$\text{Card}(A), A $	the cardinality (number of elements) of A	2
$\mathcal{P}(A)$	the power set of A (all subsets)	2
$A \times B$	the Cartesian product (set of ordered pairs)	2
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$	naturals, integers, rationals, reals	2
$n!$	n factorial	3
A_n^k	arrangements: ordered choices of k from n	3
$\binom{n}{k}$	binomial coefficient “ n choose k ”	3
$\llbracket 1, n \rrbracket$	the integers from 1 to n	3
$\sum_{k=m}^n, \prod_{k=m}^n$	sum / product over k from m to n	5
$f : A \longrightarrow B$	a function f from A to B	6
$x \longmapsto f(x)$	x maps to $f(x)$	6
$g \circ f$	g composed with f (f first, then g)	6
f^{-1}	inverse function, or preimage of a set	6
$A \sim B$	A and B are equipotent (same cardinality)	7
$(u_n), \lim_{n \rightarrow \infty} u_n$	the sequence (u_n) ; its limit	8
$\lim_{x \rightarrow a} f(x)$	the limit of f at a	9
$f'(a), f''$	first and second derivative of f	9
$\int_a^b f(x) \, dx$	the definite integral of f from a to b	10

11. Appendix: derivatives and antiderivatives

The four reference tables of elementary calculus. In the composite tables, u and v are differentiable functions of x ; in every antiderivative an arbitrary constant C is understood.

11.1. Standard derivatives

$f(x)$	$f'(x)$	differentiable on
x^α	$\alpha x^{\alpha-1}$	$(0, +\infty)$ (all of $\mathbb{R}$ when $\alpha \in \mathbb{N}$)
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$(0, +\infty)$
$\ln(x)$	$\frac{1}{x}$	$(0, +\infty)$
e^x	e^x	$\mathbb{R}$
$\cos(x)$	$-\sin(x)$	$\mathbb{R}$
$\sin(x)$	$\cos(x)$	$\mathbb{R}$
$\tan(x)$	$1 + \tan^2(x) = \frac{1}{\cos^2(x)}$	$\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi : k \in \mathbb{Z}\}$
$\arccos(x)$	$-\frac{1}{\sqrt{1-x^2}}$	$(-1, 1)$
$\arcsin(x)$	$\frac{1}{\sqrt{1-x^2}}$	$(-1, 1)$
$\arctan(x)$	$\frac{1}{1+x^2}$	$\mathbb{R}$

11.2. Composite derivatives

function	derivative
uv	$u'v + uv'$
$1/u$	$-u'/u^2$
u/v	$(u'v - uv')/v^2$
$v \circ u$	$(v' \circ u) \cdot u'$
u^α	$\alpha u' u^{\alpha-1}$
$\sqrt{u}$	$u'/(2\sqrt{u})$
$\ln(u)$	u'/u
e^u	$u'e^u$
$\cos(u)$	$-u' \sin(u)$
$\sin(u)$	$u' \cos(u)$

11.3. Standard antiderivatives

$f(x)$	an antiderivative	on
x^α ($\alpha \neq -1$)	$\frac{x^{\alpha+1}}{\alpha+1}$	$(0, +\infty)$ (all of $\mathbb{R}$ when $\alpha \in \mathbb{N}$)
$1/x$	$\ln x $	any interval avoiding 0
$1/\sqrt{x}$	$2\sqrt{x}$	$(0, +\infty)$
e^x	e^x	$\mathbb{R}$
$\sin(x)$	$-\cos(x)$	$\mathbb{R}$
$\cos(x)$	$\sin(x)$	$\mathbb{R}$
$1/\cos^2(x)$	$\tan(x)$	any interval avoiding $\frac{\pi}{2} + k\pi$
$\frac{1}{x^2+a^2}$ ($a \neq 0$)	$\frac{1}{a} \arctan\left(\frac{x}{a}\right)$	$\mathbb{R}$
$\ln(x)$	$x \ln(x) - x$	$(0, +\infty)$

11.4. Composite antiderivatives

expression	an antiderivative
$\alpha u' u^{\alpha-1}$	u^α
u'/u	$\ln u $
$-u'/u^2$	$1/u$
$u'/\sqrt{u}$	$2\sqrt{u}$
$u'e^u$	e^u